

FINAL EXAMINATION CALCULUS II - Term 2023.2

Course ID: MI1121E

Time duration: 90 mins

Attention: No document is allowed.

Question 1 (1pt): Find the curvature of $x = t + \sin 2t$, $y = 2 - \cos t$ at the point when $t = \pi$.

Question 2 (1pt): Evaluate the double integral $\iint_D 2x dx dy$ where D is defined by $-1 \leq y \leq 1$, $y^2 \leq x \leq 1$.

Question 3 (1pt): Evaluate the surface integral $\iint_S (z^2 + 1) dS$, where S is the surface $z = \sqrt{x^2 + y^2}$, $1 \leq z \leq 2$.

Question 4 (1pt): Calculate the flux of the vector field

$$\vec{F} = (3y^2 - z^2)\vec{i} + (x^2y + z^3)\vec{j} + (x^3 + y^2z)\vec{k}$$

across the outward oriented surface of the solid bounded by the cone $z = 1 - \sqrt{x^2 + y^2}$ and the xy -plane.

Question 5 (1pt): Show that $\vec{F} = (8xe^{2z} + 5y^2)\vec{i} + (10xy - 3y^2e^{2z})\vec{j} + e^{2z}(8x^2 - 2y^3)\vec{k}$ is a conservative vector field. Find a function f such that $\vec{F} = \nabla f$.

Question 6 (1pt): Evaluate $\iiint_V \sqrt{4x - x^2 - y^2 - z^2} dx dy dz$, where V is defined by $x^2 + y^2 + z^2 \leq 4x$.

Question 7 (1pt): Evaluate the following integrals:

a) $\int_0^{+\infty} \frac{x^2 \sqrt{x}}{(x^3 + 1)^2} dx.$

b) $\int_0^{+\infty} \frac{e^{-x} - 3^{-x}}{x} dx.$

Question 8 (1pt): Evaluate $\int_C (6x\sqrt{x^2 + 1} - 2xy^2) dx + (y^3 - 3x^2y^2) dy$, where C is the curve $y = \sqrt{1 - x^3}$ from $A(1; 0)$ to $B(0; 1)$.

Question 9: (1pt). Find the area of the region enclosed by $9x^2 - 4xy + y^2 = 4$.

Wishing you success in your exam!